



Web Technologies – SEF23

Homework 02- Music Library Manager

19th September, 2025 – Professor Shuja Ur Rehman Baig

Instructions:

- Student strictly must not copy code from anywhere.
- Push your code on github classroom before deadline (Homework repository will be shared later with you all).
- Viva of each homework will be conducted.
- **Deadline:** 22th September, 2025 11:59 PM

Problem Statement

You will create a simple music library management system that stores songs, manages playlists, and saves data to files and database.

What You Need to Build

1. Song Class

Create a Song class with these exact requirements:

Properties to include:

- SongId (int)
- Title (string)
- Artist (string)
- Duration (double - in minutes)
- Genre (string)
- IsLiked (bool)
- PlayCount (int)

Methods to include:

- Constructor with optional parameters (make IsLiked = false and PlayCount = 0 by default)
- PlaySong() method that increases PlayCount
- DisplaySongInfo() method



2. File Operations Classes

Create FileManager class with these methods:

For text files (using StreamWriter/StreamReader):

- SaveSongsToText(List<Song> songs, string filename)
- LoadSongsFromText(string filename) - returns List<Song>

Files to create:

- "songs.txt" (text format)
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3. Playlist Manager Class

Create PlaylistManager class with:

Properties:

- List<Song> AllSongs (It will store and manage all songs in this class – no db or files)
- string PlaylistName
- int MaxSongs

Methods:

- AddMultipleSongs(params Song[] songs)
 - CreatePlaylist(string name, int maxSongs=10)-give playlist a name, update maxsongs
 - FindSongsByGenre(string genre) - returns List<Song>
 - GetPlaylistStatistics() - shows total songs, total duration, liked songs count
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4. Database Manager Class

Create MusicDatabase class with:

Connection setup:

- Connection string for SQL Server LocalDB
- OpenConnection() method
- CloseConnection() method



Required methods using ADO.NET:

- InsertSong(Song song)
- GetAllSongs()
- GetSongCount()
- GetSongById(int id)

Database table to create:

```
CREATE TABLE Songs (  
    SongId INT PRIMARY KEY IDENTITY(1,1),  
    Title NVARCHAR(200),  
    Artist NVARCHAR(100),  
    Duration DECIMAL(4,2),  
    Genre NVARCHAR(50),  
    IsLiked BIT,  
    PlayCount INT DEFAULT 0  
);
```

Main Program – Menu Driven

Your program must provide a console menu for the user to select actions:

1. **Create Songs (at least 5)** – Add at least 5 songs with details like title, artist, and genre.
2. **Display All Songs** – Show the list of all songs currently in memory.
3. **Save Songs to Text File** – Write the songs into a text file for storage.
4. **Load Songs from Text File** – Read songs back from the text file into memory.
5. **Insert Songs into Database** – Save the songs into a database table.
6. **Display All Songs from Database** – Fetch and show all songs stored in the database.
7. **Get Song by ID (Database)** – Search and display song details by its unique ID.
8. **Get Total Song Count (Database)** – Show the number of songs currently stored in the database.



9. **Create Playlist** – Call playlist manager function of creating playlist
10. **Display Playlist Statistics** – Shows total songs, total duration, liked songs count.
11. **Find Songs by Genre** – Search songs based on their genre.
12. **Play a Song (increase PlayCount)** – Simulate playing a song and increase its play count.
13. **Exit** – Quit the program safely.